

VALOR — 8-Minute Judge Pitch Script

SIH26181 • Team CHIPSTERS • 70% Layman / 30% Technical

Total runtime target: ~7:45–8:00. Bold = say almost verbatim. [Technical beat] = the 30% layer, only go deep here.

The throughline (keep this in your head)

One story runs through all 8 slides: **a rescue worker, alone, in a disaster, with no network and no doctor nearby**. Every slide answers one question that worker's situation raises. You are not presenting 8 separate topics — you are answering 8 questions about the same person, in order.

SLIDE 1 — Title Page

Time: 0:00–0:20 | Tone: cold open, no "hello everyone"

"Picture a flood-rescue worker, standing in chest-deep water, six hours into a shift, in a place where the cell towers went down two hours ago. Nobody is monitoring her. Not a hospital, not an app, not a colleague. If her heart starts to give out from the strain, or the air turns toxic around her, she is the only sensor she has. **We built the thing that watches over her when nothing else can. This is VALOR — Team Chipsters, Problem Statement 26181.**"

(Click to Slide 1 as you say the last line — title, team, PS ID are all visible, so don't read them out loud individually.)

SLIDE 2 — Problem and Solutions

Time: 0:20–1:50 (90s) | 80% layman, 20% technical

[Layman — the problem, left side of slide]

"Why can't she just wear a Fitbit or an Apple Watch? Three reasons, and they're all on this slide. One — **her watch is blind to the world around her**. It'll count her steps, but it has no idea the air she's breathing has an AQI of 500, or that it's 48 degrees out. Two — **weather apps are blind to her**. They'll tell you it's hot in the city. They won't tell you what that heat is doing to *her* heart, right now, at street level. Three — **the cloud dies exactly when she needs it most**. Every smart health platform assumes a phone network. In a flood or a storm, that's the first thing that collapses."

[Technical beat — the 4th problem, quietly the most important one]

"And there's a fourth, quieter problem: **[Technical]** even if you kept the cloud alive, a normal processor's operating system introduces 5 to 20 milliseconds of timing jitter —

and heart rate variability, the earliest warning sign of shock, lives in a signal *smaller* than that jitter. Software literally cannot see it clearly."

[Layman — pivot to solution, right side of slide]

"So VALOR does four things differently. It fuses her **body and her environment** into one picture — pulse, oxygen, temperature, humidity, and air quality, all from the wrist. It uses a small, dedicated chip — an FPGA — that measures her heartbeat with **hardware-level precision**, not software guesswork. Its entire decision-making brain fits in **619 bytes** — small enough to run with zero cloud, zero network, forever. And it doesn't just report numbers — it predicts heat stroke, toxic-air stress, and cold shock **15 to 20 minutes before** they become emergencies."

"One line to remember: we turned a dumb sensor into an offline clinical life-saver."
(point to the blue box at the bottom of the slide)

SLIDE 3 — Technical Approach: The 4-Stage Pipeline

Time: 1:50–3:00 (70s) | 60% layman, 40% technical

[Layman — analogy first]

"So how does it actually work end to end? Think of it as a relay race with four runners, and this diagram is that relay."

[Walk the diagram left to right, matching the four stages on screen]

"**Stage 1, Sensing** — three sensors on her wrist: one reads her pulse and oxygen, one reads temperature and humidity, one reads particulate matter in the air, like a tiny smoke detector.

Stage 2, the FPGA core — this is the part that makes us different. [**Technical**] It's an 8-tap filter and a peak-detection state machine, running at 50 megahertz, giving us a 20-nanosecond timer. That's not a marketing number — it means we timestamp every single heartbeat with a precision no software running on that same chip could ever match.

Stage 3, Clinical TinyML — this is where the raw numbers become a medical judgment. We run a recognized hospital scoring system called **mNEWS2**, a signal-quality check so a runner's jostled sensor doesn't get mistaken for a heart attack, and a tiny **6 → 24 → 16 → 3** neural network making the final call.

Stage 4, Output — the verdict shows up right there on an OLED screen on her wrist, or as an edge alert. No phone required."

"Four stages, one wrist, zero network."

SLIDE 4 — Technical Approach (Stage 2): Silicon Proof

Time: 3:00–4:00 (60s) | 40% layman, 60% technical — this is your credibility slide

[Layman framing first]

"Now, anyone can draw a nice block diagram. We want to show you this isn't a concept — it's real silicon, doing real work, right now."

[Point to left image]

"This is the actual floorplan of our design synthesized onto the Renesas ForgeFPGA chip — the physical layout of logic gates that IS our heartbeat detector."

[Point to right image — the waveform]

"[**Technical**] And this is the waveform from our own testbench. Watch the red trace — that's the filtered PPG signal converging over eight samples. When it crosses the threshold, the state machine fires, and you can see `(peak_ibi_cycles)` latch a real cycle count — that number is how we get 20-nanosecond beat timing without a single floating-point operation."

[Layman close — read the blue box, this is your best analogy]

"The analogy on this slide says it best: **software CPUs are like timing a sprint with a loose, flickering stopwatch. Our FPGA is a stopwatch built into the finish line itself — 20 billionths of a second, zero lag.**"

SLIDE 5 — Software Approach (Stage 3): VALOR Clinical Intelligence

Time: 4:00-5:10 (70s) | 60% layman, 40% technical

[Layman — the headline numbers]

"So the hardware measures perfectly. What does the software *do* with that? This is where VALOR becomes a clinician, not just a sensor."

[Point to the three stat cards]

"**88.47% model accuracy** on a validation set of over 6,200 samples. **94.11% clinical concordance** — meaning when we tested it against **16,387 real ICU patient records** from a public hospital database called MIMIC-III, our triage calls agreed with real clinical outcomes 94 times out of 100. And the entire brain making that call is **619 bytes** — smaller than this sentence in computer memory."

[Technical beat — the two boxes]

"[**Technical**] On the left, that's a 6-input, 24-16-hidden-layer, 3-output network, trained with quantization-aware training so it runs as 8-bit integers instead of full floating point — that's what lets it live on a tiny edge chip instead of a phone. On the right, we didn't invent our own scoring system from scratch. We implemented **mNEWS2**, a real early-warning protocol used by the UK's Royal College of Physicians, plus **Moran's**

Physiological Strain Index and an **American Heart Association** model linking air pollution to cardiac stress. We stood on validated clinical science instead of guessing."

"One tiny model. Multiple risks. Real-world impact."

SLIDE 6 — Feasibility and Viability

Time: 5:10–6:10 (60s) | 70% layman, 30% technical

[Layman — this is the "can you actually ship this" slide]

"Great tech means nothing if it can't leave the lab. So — can this actually be built, and can it actually be afforded?"

[Top-left, technical box]

"[**Technical**] On the silicon side — our design uses zero DSP blocks and zero block RAM. That means it's pure logic, which means it's cheap to manufacture and portable — the same Verilog compiles on both AMD Xilinx and Renesas chips without rewriting a line."

[Top-right, cost box — layman, this is the number judges remember]

"On the cost side: a single prototype unit costs about ₹3,500 today. At mass production, we're targeting **under ₹1,500 per unit** — that's a **95% cost reduction** compared to commercial health monitors that run ₹25,000 to ₹40,000. And because everything runs on-device, there's **zero ongoing cloud bill** — you buy it once."

[Bottom-left, quickly]

"We've also mapped out our real engineering headaches — motion artifacts, limited FPGA capacity, pin constraints — and how we solved each one, with a proven result next to every fix."

[Bottom-right, roadmap — close with momentum]

"And we're not starting from zero. **Phase 1 is done** — hardware running, 94.11% accuracy proven on real ICU data. **Phase 2** is a 50-unit field pilot with sanitation and construction workers under real 45-degree heat. **Phase 3** takes this to commercial silicon, under 5 milliwatts of power."

SLIDE 7 — Impact and Benefits

Time: 6:10–7:10 (60s) | 85% layman, 15% technical

[Layman — zoom out from engineering to human impact]

"Step back from the circuit boards for a second. Who does this actually change?"

[Point to key metrics, bottom right — say these numbers slowly]

"**380 million-plus workers** in India alone do outdoor, high-exposure jobs — construction, sanitation, disaster response, agriculture. That's who this is for. **94.11% triage accuracy. 20-nanosecond beat timing. Just 12 parts** in the entire bill of materials."

[Social + Economic, briefly]

"Socially — because it's fully offline, it works exactly during blackouts and network failures, which is exactly when people need it most, and your health data never leaves your wrist. Economically — no subscription, no cloud bill, one-time purchase, which is what makes it viable for the workers who need it most, not just the ones who can afford a premium wearable."

[Read the quote in the middle of the slide — let it land]

"There's a line on this slide that says it better than we can: '**Same triage logic as an ICU early-warning score — running on a wrist, with no network.**'"

SLIDE 8 — Research and References

Time: 7:10-7:45 (35s) | Close strong, not a citation dump

[Layman — closing, callback to opening]

"Everything you just saw — the mNEWS2 protocol, the MIMIC-III dataset, the AHA and Moran strain models — is built on published, peer-reviewed clinical science. We didn't invent medicine. We just made it small enough to fit on a wrist and honest enough to work without a network."

[Final line — return to the rescue worker from slide 1]

"Remember the rescue worker we started with — alone, six hours in, no signal? **With VALOR on her wrist, she's never actually alone. The device is already doing the doctor's job, in real time, at 20 nanoseconds of precision, for under ₹1,500.** That's VALOR. Thank you."

Timing Cheat Sheet

Slide	Content	Time	Cumulative
1	Title / Hook	0:20	0:20
2	Problem & Solution	1:30	1:50
3	4-Stage Pipeline	1:10	3:00
4	FPGA Silicon Proof	1:00	4:00
5	Clinical TinyML	1:10	5:10
6	Feasibility & Viability	1:00	6:10
7	Impact & Benefits	1:00	7:10
8	Close	0:35	7:45

Delivery notes

- **Don't read slide text aloud verbatim anywhere** — the judges can read. Every line above adds something the slide doesn't say out loud.
- The **[Technical]** tags are your dial — if you're in a room of hardware judges, linger there an extra 10-15 seconds each; if the panel skews non-technical, compress those lines to one sentence and move on.
- The one thread to never drop: **offline, hardware-timed, and affordable**. Every slide should touch at least one of those three words.
- If a slide transition stalls (tech glitch), you already have the story memorized — you can keep talking through a black screen and it still makes sense, because the *worker's story* carries it, not the slide.